A QUASI-FINITE MORPHISM IS GENERICALLY FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. Read in full under the Lead's 12:4x rule (screening hit re-read, pb 2277B). The screening hit was the phrase 'nontrivial specialization', where 'trivial' is part of a mathematical term, not an omission; the proof is complete.

Allowed standard prerequisites (editorial).

- Local quasi-finiteness bounds the fibres above points. Lemma \ref{lemma-quasi-finite-fibre} and the surrounding discussion, used at proof lines 13230-13250. Excludes the specialisations that the proof rules out.

Source reference. Original lemma 02NW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3495–3495. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02ML: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3448–3448. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3479–3479. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01ZV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2610–2610. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02NE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3477–3477. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let X, Y be schemes. Let $f : X \to Y$ be locally of finite type. Let $\eta \in Y$ be a generic point of an irreducible component of Y. The following are equivalent:*

1. *the set $f^{-1}(\{\eta\})$ is finite,*
2. *there exist affine opens $U_i \subset X$, $i = 1, \ldots, n$ and $V \subset Y$ with $f(U_i) \subset V$, $\eta \in V$ and $f^{-1}(\{\eta\}) \subset \bigcup U_i$ such that each $f|_{U_i} : U_i \to V$ is finite.*

If f is quasi-separated, then these are also equivalent to

3. *there exist affine opens $V \subset Y$, and $U \subset X$ with $f(U) \subset V$, $\eta \in V$ and $f^{-1}(\{\eta\}) \subset U$ such that $f|_U : U \to V$ is finite.*

If f is quasi-compact and quasi-separated, then these are also equivalent to

4. *there exists an affine open $V \subset Y$, $\eta \in V$ such that $f^{-1}(V) \to V$ is finite.*

Proof. The question is local on the base. Hence we may replace Y by an affine neighbourhood of η, and we may and do assume throughout the proof below that Y is affine, say $Y = \mathrm{Spec}(R)$.

It is clear that (2) implies (1). Assume that $f^{-1}(\{\eta\}) = \{\xi_1, \ldots, \xi_n\}$ is finite. Choose affine opens $U_i \subset X$ with $\xi_i \in U_i$. By Algebra, Lemma 02ML we see that after replacing Y by a standard open in Y each of the morphisms $U_i \to Y$ is finite. In other words (2) holds.

It is clear that (3) implies (1). Assume f is quasi-separated and (1). Write $f^{-1}(\{\eta\}) = \{\xi_1, \ldots, \xi_n\}$. There are no specializations among the ξ_i by Lemma 02NG. Since each ξ_i maps to the generic point η of an irreducible component of Y, there cannot be a nontrivial specialization $\xi \rightsquigarrow \xi_i$ in X (since ξ would map to η as well). We conclude each ξ_i is a generic point of an irreducible component of X. Since Y is affine and f quasi-separated we see X is quasi-separated. By Properties, Lemma 01ZV we can find an affine open $U \subset X$ containing each ξ_i. By Algebra, Lemma 02ML we see that after replacing Y by a standard open in Y the morphisms $U \to Y$ is finite. In other words (3) holds.

It is clear that (4) implies all of (1) – (3) with no further assumptions on f. Suppose that f is quasi-compact and quasi-separated. We have to show that the equivalent conditions (1) – (3) imply (4). Let U, V be as in (3). Replace Y by V. Since f is quasi-compact and Y is quasi-compact (being affine) we see that X is quasi-compact. Hence $Z = X \setminus U$ is quasi-compact, hence the morphism $f|_Z : Z \to Y$ is quasi-compact. By construction of Z we see that $\eta \notin f(Z)$. Hence by Lemma 02NE we see that there exists an affine open neighbourhood V' of η in Y such that $f^{-1}(V') \cap Z = \emptyset$. Then we have $f^{-1}(V') \subset U$ and this means that $f^{-1}(V') \to V'$ is finite. □